

SEMINAR 4

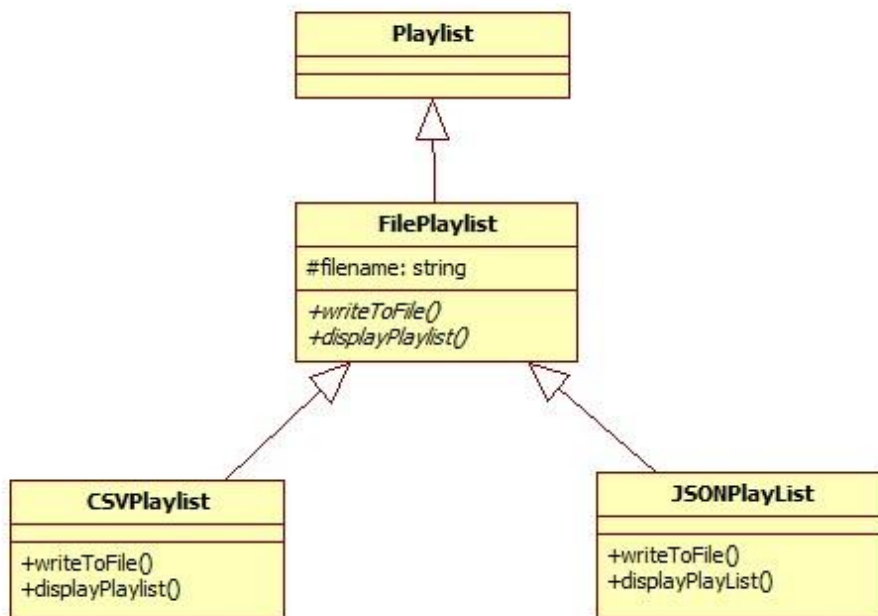
1. OBJECTIVES

- Work with input/output streams, define custom extraction/insertion operators.
- Read/write data from/to files.
- Define and use exception classes and hierarchies.

2. PROBLEM STATEMENT

Extend your playlist application to offer data persistence. The songs in the repository will be read from/written to a text file and the application will provide functionalities for storing the playlist data to a CSV/JSON file. Furthermore, the application will have to use exceptions in order to signal errors.

Implement the classes **FilePlaylist**, **CSVPlaylist** and **JSONPlaylist**, such that when a user chooses to save his/her playlist to a specific type of file, the application stores the playlist correctly and opens it using the suitable application (Chrome/Notepad for JSON and Notepad/Excel/OpenOffice Calc for CSV).



Songs JSON example:

```
{ "songs": [  
  {  
    "artist": "Pink Floyd",  
    "title": "Comfortably numb",  
    "minutes": 6,  
    "seconds": 53,  
    "link": "https://www.youtube.com/watch?v=_FrOQC-zEog"  
  },  
  {  
    "artist": "Hozier",  
    "title": "Take me to church ",  
    "minutes": 4,  
    "seconds": 2,  
    "link":  
    https://www.youtube.com/watch?v=MYSVMgRr6pw&index=30&list=PLb5DqBOB\_Gn7CfN91JA139ZWX4IwqcBBz  
  }  
]}
```